Rohan Shah

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Education

Purdue University West Lafayette, IN

B.S. in Computer Science, Artificial Intelligence Track *Expected Graduation: 2026*

- *Relevant Coursework*: Data Structures and Algorithms, Machine Learning, Artificial Intelligence, Computer Architecture, Systems Programming, Object-Oriented Programming, Analysis of Algorithms

Experience

Software Engineer Intern May 2025 – August 2025

Visa *Austin, TX*

- Developed an agentic cybersecurity reporting platform using LangGraph and OpenAI Agents, enabling users to generate complex security reports and dynamic graph visualizations through natural language queries.
- Automated report generation workflows, reducing the time to produce detailed security reports from weeks of developer effort to just minutes, and significantly improved analyst productivity and resource allocation.
- Delivered a streamlined executive access pipeline, allowing leadership to instantly obtain tailored security metrics and visualizations, and accelerated agentic data-driven decision making.
- Created two Model Context Protocol (MCP) servers used by AI agents, one to enable interactions with MongoDB and another to support dynamic graph generation, enhancing data accessibility and visualization.
- Built an AI Semantic Cache library leveraging cosine similarity and providing autocomplete functionality, which reduced end-to-end query response time by more than 3x, improved accuracy, and lowered LLM token usage.
- Generalized and inner-sourced the AI Semantic Cache as an internal tool, which was adopted across multiple production projects and teams at Visa.

Software Engineer Intern June 2024 – August 2024

ThetaRho *San Jose, CA*

- Designed and built a developer dashboard that streamlined API and container testing, provided live application monitoring, and simplified deployment workflows.
- Developed, documented, and tested REST API endpoints to enhance the platform's reliability and usability for internal and external developers.
- Implemented DevOps solutions including CI/CD pipelines, containerization, and infrastructure automation to optimize development and deployment processes.

Projects

Danny.ai | *FastAPI, React, PostgreSQL, OpenAI API, Gemini, ElevenLabs, LangChain, Supabase* 2025

- Developed a Generative AI-powered project manager and dashboard for home remodeling, awarded "Most Promising AI Startup Idea" at Purdue's largest AI hackathon (300+ students, 50+ teams).
- Implemented a context-aware RAG chatbot for project Q&A, real-time budget/timeline tracking, generative interior design previews with Gemini, and automated supplier calls using ElevenLabs.
- Integrated FastAPI-based REST APIs backed by PostgreSQL with LLMs, Gemini for AI image generation, and ElevenLabs for voice automation, delivering an intelligent, all-in-one platform for construction management.
- Created a React dashboard supporting live project status, interactive material sourcing, and seamless user communication.

algowar | *Spring Boot, Next.js, PostgreSQL, AWS Cognito/RDS/S3/ECR/Lambda/EC2, Docker* 2025

- Developed a full-stack web app to gamify Leetcode-style coding prep through competitive head-to-head matches.
- Built a custom REST API with Spring Boot and PostgreSQL hosted on AWS RDS, and implemented secure user authentication with AWS Cognito.
- Designed a modern Next.js front-end supporting real-time gameplay, a full social system, leaderboard, and profile pages, and enhanced gameplay with containerized code execution, matchmaking, chat, and solo puzzles.

Technical Skills

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, NoSQL
Frameworks: LangChain, LangGraph, Spring Boot, FastAPI, Flask, React, Next.js, Node.js
Libraries: OpenAI, Anthropic, pandas, NumPy, matplotlib, Selenium, BeautifulSoup
Databases: PostgreSQL, MongoDB
Developer Tools: Linux, Git, AWS, Docker, Postman